

Aim:

Writing a MATLAB program to plot exactly 10 cycles of a cosine signal with a frequency of 500 Hz, maximum amplitude of 5V and constant phase of $\pi / 4$

Theory:

We know that a cosine signal function is a periodic waveform described by

$$x(t) = A \cdot \cos(2\pi f t + \phi)$$

Where A is the amplitude, f is the frequency in Hertz, t is time in seconds, and ϕ is the phase shift in radians. Here, $A = 5V$ sets the value of the maximum voltage, $f = 500$ Hz determines the oscillation rate, and $\phi = \pi/4$ shifts the waveform right by 1/4 of a cycle. Since we get 500 Hz as the oscillation rate $T = 1/f = 2 \cdot 10^{-3}$ seconds.

In MATLAB, we will emulate a continuous signal by creating small increments in time (Δt) at the magnitude of 10^{-6} .

MATLAB Code:

```
clc;
clear all; % clear
close all;

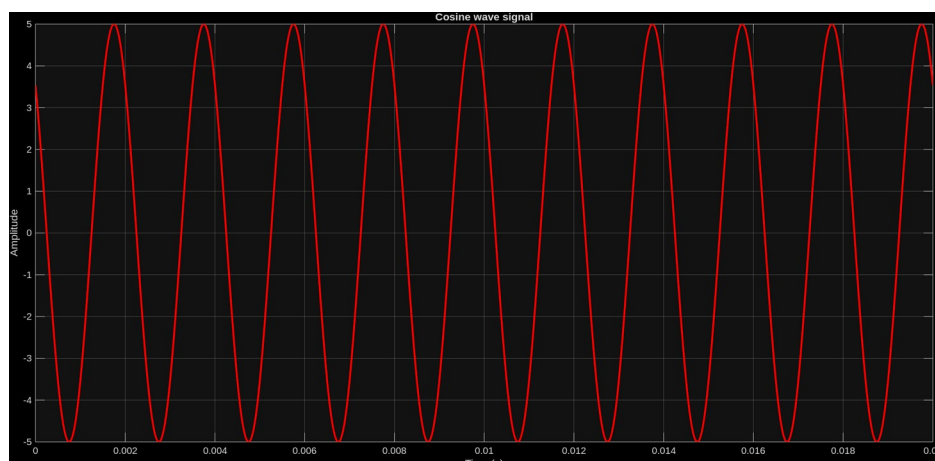
% cos Wave Signal

f = 500
T = 1/f; % Declare Variables
t = 0:10^-6:10*T;

signal = 5*cos(2*pi*f*t + (pi/4)); % Obtain Signal

plot(t,signal, 'r', LineWidth=2) % Plot signal
xlabel('Time (s)')
ylabel('Amplitude')
title('Cosine wave signal')
grid on;
```

Results:



Conculsion:

From this experiment we were able to plot and visualize 10 cycles of a cosine signal with phase difference of $\pi/4$ and a maximum amplitude of 5V in MATLAB. In MATLAB, we were able to make a smooth visualization via using a small time sample to prevent aliasing.